

1 P1 C5 — $d + 2$ lines in $\mathbb{R}^d$

For every $d \geq 2$, let $N = d + 2$ lines through the origin in $\mathbb{R}^d$, with repetitions allowed. For unit representatives x_i , write $\theta(x, y) = \arccos |\langle x, y \rangle|$, $\delta(x, y) = \pi/2 - \theta(x, y) = \arcsin |\langle x, y \rangle|$, $S = \sum_{i < j} \theta(x_i, x_j)$, and $D = \sum_{i < j} \delta(x_i, x_j)$. We prove the sharp inequality

$$S \leq \left(\binom{N}{2} - 2 \right) \frac{\pi}{2}, \quad \text{equivalently} \quad D \geq \pi.$$

The argument includes the auxiliary sparse-maximizer reduction, small-cycle inequalities, and projection induction needed for a standalone proof. Equality holds for the coordinate axes with two axes repeated.

1.1 1. The auxiliary bound and sparse maximizers

For every $d \geq 1$ and $d + 1$ lines, the C3 statement asks for

$$S \leq \left(\binom{d+1}{2} - 1 \right) \frac{\pi}{2}.$$

Equivalently, their total deficiency is at least $\pi/2$. We prove the following auxiliary bound for every $m \geq 2$: any m unit vectors whose span has dimension at most $m - 1$ have total deficiency at least $\pi/2$. Embed their span into $\mathbb{R}^{m-1}$; then set $N = m$ and $d = m - 1$ in the argument below.

1.1.1 Sparse-maximizer argument

For fixed N, d , the angle sum attains a maximum on the compact product of N unit spheres. Among maximizers choose one with the largest number of orthogonal pairs.

For a vector x with at least one nonorthogonal neighbor, let W span its orthogonal neighbors. If $\dim W < d - 1$, choose a unit y orthogonal to W and to x . Move x along $x(t) = x \cos t + y \sin t$. Every existing orthogonality is preserved. For any other vector z not orthogonal to x , its inner product is $A \cos t + B \sin t$, with $A \neq 0$. On the interval between its consecutive zeros containing $t = 0$, its line angle is convex. To check this, write the signed inner product as $r \cos(t - t_0) > 0$, $0 < r \leq 1$, and put $u = t - t_0$. For $r < 1$,

$$\frac{d^2}{du^2} \arccos(r \cos u) = \frac{r(1 - r^2) \cos u}{(1 - r^2 \cos^2 u)^{3/2}} \geq 0.$$

For $r = 1$ the angle is an absolute-value function on that interval, also convex. Intersect the closed intervals bounded by the nearest zeros for all nonorthogonal neighbors. The total angle sum is convex there by continuity at the endpoints, and has its global maximum at the interior point 0, so it is constant. At an endpoint at least one additional pair becomes orthogonal, while no orthogonal pair is lost. This contradicts the choice of maximizer.

Here $N > d$, so at least one pair is nonorthogonal. If the configuration spanned dimension $r < d$, an endpoint x of such a pair would have $\dim W \leq r - 1 < d - 1$, contradicting the preceding rotation argument. Thus the configuration spans $\mathbb{R}^d$. For a vector orthogonal to every other vector, those other vectors consequently span its entire orthogonal complement; its orthogonal-neighbor span has dimension $d - 1$ too. Every vector therefore has at least $d - 1$

linearly independent orthogonal neighbors. Its nonorthogonality graph (an edge means nonzero inner product) has maximum degree at most $N - d$.

For $N = d + 1$, this graph consists of edges and isolated vertices. The Gram matrix has nullity one. Some two-vertex block must have rank one, hence its two lines coincide and contribute deficiency $\pi/2$. This proves the auxiliary deficiency bound for an angle maximizer. Every other configuration has angle sum no greater than that maximizer, so its deficiency is no smaller. Taking $m = d + 1$ proves C3. Equality is attained by taking d mutually orthogonal coordinate axes and repeating one of them: the only positive deficiency is $\pi/2$. This also covers $d = 1$ and repeated lines.

1.2 2. Scalar and small-cycle lemmas

The C4 statement asks for $S \leq 4\pi$ for five lines in $\mathbb{R}^3$ and $S \leq 13\pi/2$ for six lines in $\mathbb{R}^4$. In either case, with $D = \sum_{i < j} \delta(x_i, x_j)$, this is equivalent to $D \geq \pi$. We use the sparse-maximizer argument above with $N = d + 2$.

1.2.1 Sparse graph and small components

For $N = d + 2$, the sparse graph consists of paths and cycles. Gram blocks belonging to different components are orthogonal. An irreducible path block has nullity at most one: the first row and the tridiagonal recurrence determine every kernel coordinate from the first coordinate. An irreducible cycle block has nullity at most two: two adjacent coordinates determine all others. The whole matrix has nullity exactly two. If two components are singular, the C3 auxiliary bound gives deficiency at least $\pi/2$ in each, proving the target. Otherwise one cycle has nullity two, and all other components are nonsingular.

For $N \leq 6$ only cycles of lengths 3, 4, 5, 6 remain.

A three-cycle of nullity two has rank one: all three lines coincide and its deficiency is $3\pi/2$. A four-cycle of nullity two has rank two: its opposite pairs are orthonormal bases of the same plane. For every unit planar vector, its acute angles to an orthogonal pair sum to $\pi/2$. Consequently the four edge deficiencies sum to π .

1.2.2 Two elementary analytic inequalities

For $p, q \in [0, \pi/2]$, put

$$h = \arccos(\cos p \cos q),$$

$$E(p, q) = \arcsin\left(\frac{\sin p \sin q}{1 + \cos p \cos q}\right).$$

Then

$$h + E(p, q) \leq p + q. \tag{B}$$

Here is a direct proof, with no geometric theorem required. Set $A = \cos p$, $B = \cos q$, $r = \sin p \sin q$, $u = AB$, and $L = \sqrt{1 - u^2}$. The identity

$$(1 + u)^2 - r^2 = (A + B)^2.$$

gives $\cos E = (A + B)/(1 + u)$. Therefore

$$(1 + u)[\cos(h + E) - \cos(p + q)] = -u(1 - A)(1 - B) + r(1 + u - L) \geq 0.$$

Indeed $L \leq 1$ and $r \geq (1 - A)(1 - B)$, since $\sin t \geq 1 - \cos t$ on $[0, \pi/2]$. Both arguments compared lie in $[0, \pi]$, where cosine is decreasing. This proves (B).

A second inequality, for $T, p, q \in [0, \pi/2]$, is

$$\arcsin(\sin p \sin q) \leq \arccos(\cos T \cos p) + \arccos(\sin T \cos q) - \frac{\pi}{2}. \quad (\text{C})$$

Set $X = \cos T \cos p$, $Y = \sin T \cos q$, and $Z = \sin p \sin q$. Direct expansion gives

$$X^2 + Y^2 + Z^2 + 2XYZ - 1 = -(\cos T \sin p \cos q - \sin T \sin q \cos p)^2 \leq 0.$$

For nonnegative $X, Y, Z \leq 1$ this implies

$$\sqrt{1 - X^2} \sqrt{1 - Y^2} \geq Z + XY.$$

hence $\cos(\arccos X + \arccos Y) \leq -Z$. Since $\arccos X + \arccos Y \in [0, \pi]$, cosine monotonicity gives their sum at least $\pi - \arccos Z$. This is (C).

1.2.3 The five-cycle

Its Gram rank is three. Take $x_1 = e_1, x_3 = e_2$, and choose signs of coordinates so that, in absolute value,

$$x_4 = (0, c, \sqrt{1 - c^2}), \quad x_5 = (a, 0, \sqrt{1 - a^2}).$$

where $0 < a, c < 1$. Possible remaining signs do not affect any formula below. Since x_2 is orthogonal to x_4 and x_5 , their cross product determines it up to sign. With $L = \sqrt{a^2 + c^2 - a^2 c^2}$, the absolute edge inner products are

$$\frac{c\sqrt{1 - a^2}}{L}, \quad \frac{a\sqrt{1 - c^2}}{L}, \quad c, \quad \sqrt{1 - a^2}\sqrt{1 - c^2}, \quad a.$$

Write $a = \sin p, c = \sin q$. If the first two deficiencies are U, V , then

$$\begin{aligned} \cos(U + V) &= \frac{\sin p \sin q}{1 + \cos p \cos q}, \\ \sin(U + V) &= \frac{\cos p + \cos q}{1 + \cos p \cos q}. \end{aligned}$$

Thus $U + V = \pi/2 - E(p, q)$. The middle product has deficiency $\pi/2 - \arccos(\cos p \cos q)$. The total edge deficiency is therefore

$$\pi + p + q - \arccos(\cos p \cos q) - E(p, q) \geq \pi.$$

by (B). The expressions extend continuously to boundary cases; alternatively those cases break the cycle and were already handled. This completes the five-line $\mathbb{R}^3$ bound.

1.2.4 Removing a vertex from the six-cycle

The six-cycle has Gram rank four. Remove x_6 and project x_1 and x_5 orthogonally onto $x_6^\perp$, normalizing the results. The other three vectors already lie in that hyperplane. The resulting five lines lie in $\mathbb{R}^3$.

Only four original deficiencies and three projected deficiencies need comparison. Relabel the local path $(x_2, x_1, x_6, x_5, x_4)$ as $(v_1, v_2, v_3, v_4, v_5)$, and choose signs making its consecutive inner products a, b, c, d positive. The odd vectors v_1, v_3, v_5 are orthonormal. In an orthonormal basis extending them,

$$v_2 = (a, b, 0, u), \quad v_4 = (0, c, d, v), \quad uv = -bc.$$

All $a, b, c, d, |u|, |v|$ are positive for an irreducible six-cycle. Choose the fourth basis direction so $u > 0$ and $v < 0$. Consequently there are $P, Q, T \in (0, \pi/2)$ with

$$\begin{aligned} a &= \cos P, & b &= \sin P \cos T, & u &= \sin P \sin T, \\ d &= \cos Q, & c &= \sin Q \sin T, & v &= -\sin Q \cos T. \end{aligned}$$

Let $B = \arcsin b, C = \arcsin c$. Let p, q be the acute angles of the normalized projections of v_2, v_4 to v_1, v_5 respectively. Then

$$\begin{aligned} \cos P &= \cos B \cos p, & \cos Q &= \cos C \cos q, \\ \tan B &= \sin p \cot T, & \tan C &= \sin q \tan T. \end{aligned}$$

The new connecting inner product has absolute value $\sin p \sin q$. Hence the original local deficiency is $\pi - P - Q + B + C$, whereas the projected local deficiency is $\pi - p - q + \arcsin(\sin p \sin q)$.

By (B),

$$B + p - P \geq E(B, p), \quad C + q - Q \geq E(C, q).$$

The tangent relations imply the following exact identities:

$$\begin{aligned} E(B, p) &= \arccos(\cos T \cos p) - T, \\ E(C, q) &= \arccos(\sin T \cos q) - \left(\frac{\pi}{2} - T\right). \end{aligned}$$

For completeness the first identity follows by substituting $\sin B \sin T = \cos B \sin p \cos T$ into the cosine addition formula for $E(B, p) + T$; the result is $\cos T \cos p$. Both angles are in $[0, \pi]$, making cosine injective. The second is the same computation with T replaced by $\pi/2 - T$.

Inequality (C) now gives

$$(B + p - P) + (C + q - Q) \geq \arcsin(\sin p \sin q).$$

Thus projection does not increase total deficiency. The projected five-line configuration has deficiency at least π by the result just proved. The original six-cycle therefore also has deficiency at least π .

Combining all component cases proves both C4 bounds for a sparse angle maximizer. Every other configuration has no larger angle sum, so the bounds hold for all five- and six-line configurations.

Equality is attained by coordinate axes with two axes repeated: multiplicities $(2, 2, 1)$ in $\mathbb{R}^3$ and $(2, 2, 1, 1)$ in $\mathbb{R}^4$.

1.3 3. The all-dimensional bound

1.3.1 3.1 The scalar projection inequality in its general form

For acute B, U define

$$g(B, U) = B + U - \arccos(\cos B \cos U).$$

For $B, C, U, V \in (0, \pi/2)$, suppose

$$\tan B \tan C \leq \sin U \sin V.$$

Then

$$\arcsin(\tan B \tan C) \leq g(B, U) + g(C, V). \quad (1)$$

The arcsine argument is in $(0, 1)$, since $\sin U \sin V < 1$. Put

$$U_0 = \arcsin\left(\frac{\tan B \tan C}{\sin V}\right).$$

The hypothesis and sine monotonicity on $[0, \pi/2]$ give $0 < U_0 \leq U < \pi/2$. Choose $T \in (0, \pi/2)$ with

$$\tan T = \frac{\sin U_0}{\tan B} = \frac{\tan C}{\sin V}.$$

Thus

$$\begin{aligned} \sin B \sin T &= \cos B \sin U_0 \cos T, \\ \sin C \cos T &= \cos C \sin V \sin T. \end{aligned}$$

The equality-case projection inequality proved in Section 2's six-cycle argument gives

$$\arcsin(\sin U_0 \sin V) \leq g(B, U_0) + g(C, V).$$

Its left side is $\arcsin(\tan B \tan C)$. It remains to compare U_0 with U . For fixed $B \in (0, \pi/2)$,

$$\frac{\partial g(B, U)}{\partial U} = 1 - \frac{\cos B \sin U}{\sqrt{1 - \cos^2 B \cos^2 U}} \geq 0.$$

The denominator is positive, and its square minus $(\cos B \sin U)^2$ equals $\sin^2 B$. Consequently $g(B, U_0) \leq g(B, U)$, proving (1). This monotonicity extension is written only; the equality-case inequality is Lean-verified.

1.3.2 3.2 Projection of a cycle in any dimension

Consider an irreducible nonorthogonality cycle of length $k \geq 6$ and rank $k - 2$. Remove one vertex and project its two neighbors onto its orthogonal complement, normalizing both. All remaining vertices were already orthogonal to the removed vertex.

Write the five-vertex path around the removed vertex as $(v_1, v_2, v_3, v_4, v_5)$, with v_3 removed. Orient the representatives along this path so that the four consecutive inner products a, b, c, d are positive. Because $k \geq 6$, the odd vectors v_1, v_3, v_5 are orthonormal and $v_2 \perp v_4$. Decompose

$$\begin{aligned} v_2 &= av_1 + bv_3 + u, \\ v_4 &= cv_3 + dv_5 + v, \end{aligned}$$

where u, v are orthogonal to the span of v_1, v_3, v_5 . Then

$$\begin{aligned} a^2 + b^2 + \|u\|^2 &= 1, \\ c^2 + d^2 + \|v\|^2 &= 1, \\ \langle u, v \rangle &= -bc. \end{aligned}$$

Cauchy–Schwarz yields $bc \leq \|u\|\|v\|$. Since $a, b, c, d > 0$, the residual vectors are nonzero and

$$A = \sqrt{1 - b^2} > 0, \quad Q = \sqrt{1 - c^2} > 0.$$

Set

$$\begin{aligned} B &= \arcsin b, & C &= \arcsin c, \\ \cos U &= a/A, & \sin U &= \|u\|/A, \\ \cos V &= d/Q, & \sin V &= \|v\|/Q. \end{aligned}$$

These specify angles $B, C, U, V \in (0, \pi/2)$. Moreover,

$$\tan B \tan C = \frac{bc}{AQ} \leq \frac{\|u\|\|v\|}{AQ} = \sin U \sin V.$$

Only four old deficiencies and three new deficiencies change. The original local contribution is

$$\arcsin a + B + C + \arcsin d = \pi - \arccos(\cos B \cos U) - \arccos(\cos C \cos V) + B + C.$$

The normalized projections of v_2, v_4 have inner products a/A with v_1 , d/Q with v_5 , and $-bc/(AQ)$ with each other. Their local contribution is therefore

$$\pi - U - V + \arcsin(\tan B \tan C).$$

Inequality (1) says that the new contribution is at most the old contribution. Every other surviving inner product is unchanged: vectors outside the two projected neighbors are orthogonal to v_3 , and all nonincident products with either projected neighbor were zero. The removed vertex had nonzero products only with those two neighbors. Thus projection does not increase total deficiency.

The original cycle spans a space of dimension $k - 2$. After projection, the $k - 1$ remaining lines lie in a space of dimension at most $k - 3$. No assertion about their new Gram rank or graph structure is needed.

1.3.3 3.3 Induction and component accounting

We prove the desired bound by strong induction on $N \geq 4$. The sparse-maximizer argument in Section 1 applies in every finite dimension: some angle maximizer has at most $N - d = 2$ nonorthogonal neighbors per line. The corresponding general geometric reduction is also Lean-verified as `C4SparseConfiguration.exists_degree_two_maximizer`.

Its nonorthogonality graph has only path and cycle components. The Gram matrix is block diagonal across these components. Its rank is at most $d = N - 2$, so its total nullity is at least two; full spanning is unnecessary here. An irreducible path block has nullity at most one; an irreducible cycle block has nullity at most two. These matrix bounds are Lean-verified under their explicit ordered-band hypotheses in `C4MatrixCorank.lean`.

If at least two components are singular, each contributes deficiency at least $\pi/2$ by the general auxiliary bound proved in Section 1. Their total already gives $D \geq \pi$. If exactly one component is singular, it must be a cycle of nullity two, of some length $k \leq N$, and its rank is $k - 2$.

The small cycles are settled as follows:

- $k = 3$: rank one forces all three lines to coincide, giving deficiency $3\pi/2$.
- $k = 4$: the two opposite orthogonal pairs are bases of one plane; the four edge deficiencies sum to π .
- $k = 5$: the explicit pentagon inequality in Section 2 gives deficiency at least π . Its scalar coordinate bound is Lean-verified in `C4FiveCycle.lean`.

These cases include the $N = 4$ and $N = 5$ bases. If $k \geq 6$, apply the general projection of Section 3.2. The resulting $k - 1$ lines lie in dimension at most $k - 3$ and $k - 1 < N$. Embed their span into $\mathbb{R}^{k-3}$ if necessary. The induction hypothesis gives projected deficiency at least π , and projection did not increase deficiency. The original cycle therefore contributes at least π too. Every other component contributes a nonnegative amount.

This proves $D \geq \pi$ for the sparse angle maximizer. Since this configuration maximizes S and $D = \binom{N}{2}\pi/2 - S$, every other configuration has deficiency at least as large. Hence the C5 angle bound holds for all configurations. Equality is attained by coordinate axes with multiplicities $(2, 2, 1, \dots, 1)$, which have exactly two coincident pairs.

1.4 Verification boundary

This is a complete written mathematical proof. Selected analytic, sparse-geometric, and matrix lemmas have Lean-checked companions; the full general $d + 2$ theorem and the final projection induction are not formalized in Lean.